

Designing Extended Reality Guidance for Physical Caregiving Tasks

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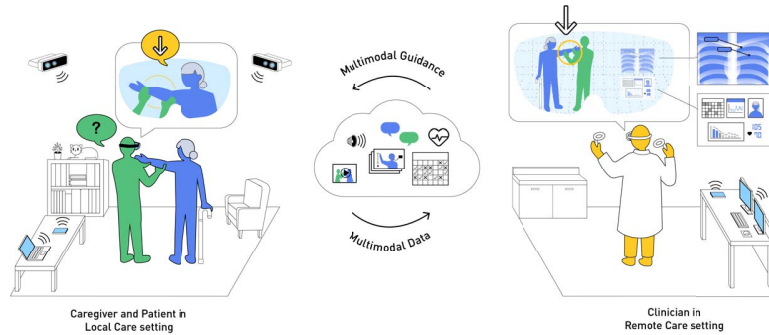


Figure 1: XR Care use case. A caregiver assists a patient with essential rehabilitation exercises, guided by a remote clinician.

ABSTRACT

In this short paper we explore the opportunities and challenges of designing XR technologies to support the collaborative work between family caregivers and clinicians as they attend to the physical care needs of patients in the home setting.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques

1 INTRODUCTION

Home care is essential for recovery and improved quality of life for individuals with short- and long-term health needs [5]. Frequently, family members provide this essential at-home assistance [6, 7, 10] with the AARP estimating that more than 1 in 5 Americans have provided care to an adult or child with special needs in the past year [1]. Family caregivers support a broad range of in-home tasks from activities of daily living to emotional support. They help patients carry out routine health assessments (e.g., take vital signs, assess swelling), clinical follow-up tasks (e.g., change bandages, use medical oxygen), and assess and assist with mobility (e.g., transfer to a sitting position, assist with range of motion exercises).

Technological interventions such as wearables, IoT, Remote Patient Monitoring, and robotic devices will increasingly assist caregivers to autonomously conduct a growing number of tasks. However, informal caregivers in particular will continue to require expert guidance to cope with the personalized and dynamic nature of patients' care needs. Hence, remote guidance for family caregivers as they conduct physical care tasks will remain a long-term need; both to address immediate execution of tasks and contribute to caregiver training. Without such guidance, caregivers may avoid needed tasks, leading to poor adherence and consequentially increased and avoidable burdens on the patient, their family and the healthcare system.

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Ongoing research in the Extended Reality (XR) community is advancing the use of AR and VR to support collaborative physical work tasks [2, 17, 32, 39]. In this short paper we explore the opportunities and challenges of designing XR technologies to support the collaborative work between family caregivers and clinicians as they attend to the physical care needs of patients in the home setting. Our multidisciplinary approach integrates methods from Human-computer Interaction (HCI), Interaction Design, and Interaction Theory to lay the foundation for future research and development of diverse XR applications that could ultimately transform the delivery of care at home.

2 RESEARCH VISION

XR Care looks beyond patient monitoring to address the myriad *physical caregiving tasks* that occur in the home. We aspire to advance the state of the art in XR technologies to the point that caregivers can receive detailed, interactive, multimodal guidance from remote experts across diverse and adaptive care tasks.

To date, XR for medicine has been studied largely in relatively controlled environments with highly-trained users (e.g., assisted surgery), or for activities that patients do for themselves (e.g., physical therapy avatars, smart mirrors, and multimedia conversational agents). Our research will extend these advances to home care settings where family caregivers are essential yet untrained actors, remote experts are integral advisors, and patients are aware and considered participants. XR Care will augment these human actors to perform tasks that fully automated techniques cannot, or are not desired to, perform [41, 42].

Figure 1 illustrates our vision. In this scenario, a caregiver is assisting a patient with daily, progressively difficult rehabilitation exercises after a mastectomy, but the patient stops making progress due to increased pain. XR Care systems enable a remote clinician to assess the problem and offer new guidance to unlock progress. Specifically, the local caregiver's augmented reality (AR) capabilities allow the local caregiver to show the remote clinician precisely which movements are painful, and to expose details of the incision sites. The remote clinician's virtual reality (VR) capabilities allow her to view high-fidelity details of the patients bandages and drainage areas, even as the patient moves through their exercises. The remote clinician can also point, gesture, annotate, and otherwise visually demonstrate to the caregiver how to adjust the patients posture, or identify if the surgical incision site needs additional medical attention. The same system can record and replay home care procedures

to support the caregiver in repeating the task when the clinician is not present; and in the farther future, intelligent agents may be developed to provide adapted verbal and visual guidance. This research agenda pursues the essential multi-stakeholder research, interaction technique development, and experimental studies on which to base these envisioned future advances.

This is an essential and well-timed area for experimentation. State-of-the-art XR systems are now capable of supporting interaction design and HCI studies of care collaboration at a distance. The family caregiving setting poses significantly different opportunities, challenges, and constraints than the more prevalent entertainment and industrial applications that have shaped XR technologies to date. And the opportunity for XR to support family caregiving will benefit patients, families, and healthcare service delivery.

Although XR Care will ultimately be adaptive and versatile enough to meet the needs of patients and caregivers facing a wide range of health challenges, we draw our examples from two specific contexts: caring for patients with *heart failure* and *cancer* [7, 25, 28, 38]. Both of these conditions are pressing public health problems, have a well-established need for home-based care, and involve a range of physical caregiving tasks. For example, heart failure patients require frequent checks for declining heart function, management of comorbidities such as diabetes, and ongoing assistance to encourage safe physical activity; and cancer patients have needs ranging from wound care, to IV administration, and physical therapy. Our research focuses on tasks that are already intended for home administration but for which family caregivers often lack familiarity and confidence and thus need guidance from visiting nurses, additional clinic visits, or otherwise risk poor management or adherence. Within these contexts, XR Care will assist with a range of care tasks involving both medical assessment and intervention at home. *Assessments* characterize the patient's evolving health status and needs: Is the patient's heart failure-related leg swelling worsening? Is a surgical incision site of a recovering cancer patient showing signs of infection? Is the patient ready to start walking without assistance? *Interventions*, such as rehab exercises, medication administration, or wound care, need to be adapted to each patient as their conditions change.

3 XR CARE INTERACTION TECHNIQUES

Our envisioned XR Care system supports a local worker (the caregiver) who is co-present in a shared physical space with the task subject/participant (the patient), and a remote expert (clinician). As with all remote collaboration, the communication space is virtual and participants must rely on technical mediation for communication. Augmenting communication cues is both more critical and more challenging for physical task collaboration, relative to strictly digital-task collaboration.

An important aspect of interactions in envisioned XR Care systems is that they involve three human parties: clinicians, caregivers, and patients. Whereas, remote guidance activities in parts assembly [9, 16, 20, 31, 43], sports coaching [19], spatial navigation [35], maintenance [3, 13, 14], and home repair tasks [11, 15] involve physical tasks on inanimate objects, the subject of physical activities in most caregiving tasks is a human patient. Moreover, in contrast to other medical remote guidance activities [30, 37, 40], caregiving tasks in the home are conducted by untrained local caregivers in uncontrolled environments, and involve physically aware and active patients. As a result, XR Care presents novel challenges and opportunities that call for the development and evaluation of new interaction techniques.

XR Care System Description. We have implemented an early prototype of an XR Care system using head-mounted display technologies with audio and video pass-through for augmented reality. It also supports live recognition, visual annotation, data recording and collection of activities. In this setup, the *local caregiver* is



Figure 2: XR Care prototype showing pointing, gestures, gaze, annotations. *Top left*: local view of caregiver through AR headset. *Bottom left*: local site of caregiver & patient. *Top right*: remote view of clinician through VR headset. *Bottom right*: remote site of clinician.

equipped with an optical see-through head-mounted display (OST HMD). This augmented reality device overlays virtual content on top of the real environment, in the field of view of the caregiver. The virtual content includes annotations and gesture visualizations from the remote clinician. The OST HMD is untethered, and features hand tracking capabilities that allow gestural control operations of the system, without the need for further input devices which would complicate the physical tasks the caregivers are performing. The *local task environment* is also captured with depth sensors. This synchronized depth and color data is then reconstructed into a real-time 3D representation of the task environment, which includes a live 3D representation of the caregiver and the patient. The *the patient* is the task subject, an observer, and a participant in the communication. The patient is not burdened with wearing or using any additional devices; their actions and communication are captured through sensing of the local task environment. The patient's role, reactions, and perspectives are considered and evaluated throughout our studies. The *remote clinician* uses a virtual reality headset to interact with a real-time 3D reconstruction of the local task environment. This enables the clinician to freely navigate around the virtual representation of the local scene, exploring the task environment from independent viewing angles. Furthermore, the pose of the headset and handheld input devices are transmitted to the local caregiver's headset, which renders a representation of that data based on the interaction primitives, allowing for the transmission of non-verbal cues such as gaze and gestures.

Figure 2 shows screen captures and snapshots from this early prototype. The XR Care system is defined around task rather than technology. While the current XR Care system is based on using OST HMD, the system architecture is designed to be modular and to accommodate different input and output devices, iterative refinement, and new features. For example, we will actively examine whether and how competing AR technologies such as room-level projection or mobile phone-based AR might equally or better support the clinician guided care. This range of deployment options may narrow down to some optimal configuration, or may be maintained as different configurations that enable more widespread adoption.

Interaction Techniques. The conversational grounding process for collaborative physical tasks has been described as consisting of three phases: (1) Reference an object or location in the task space; (2) Transmission of procedural instructions for the task; and (3) Confirmation of comprehension and task progress [8, 22, 23]. Co-located collaboration in these types of tasks is supported by communication cues such as speech, gaze, and gestures [21]. Fussell et al. developed pointing and representational gestures to aid this conversational grounding in remote collaboration on physical tasks, using surrogate pointing through a cursor and drawn annotations in the video space

to aid grounding [9]. Giusti et al., studying groups repairing Lego models and punctured bicycle tubes, affirmed these phases in the grounding process, but also noted that remote collaborators create different “stage setups” for face-to-face conversation, presenting to the collaborator, comparing (two-way presentations), workspace viewing, pausing and sharing documents [11].

This prior work provides a rich set of XR interaction techniques to draw upon. However, much of this work addressed two-party interactions on passive objects (e.g., static objects or sedated patients.) XR Care tri-partite scenarios raise new challenges that call for reexamination of established XR interaction techniques. For example: (a) the caregiver and clinician divide attention between the patient, the task, and each other, complicating conversational grounding; (b) the grounding and staging occur with respect to the patient, and annotations must follow the patient’s movements; and (c) any gestural manipulation of XR Care software by the caregiver needs to be done in a way that is socially appropriate to both the remote clinician and the patient. Below we describe interaction techniques to address these challenges.

Interaction Primitives. Remote collaboration on physical tasks is inherently challenging due to the disconnect between the communication space, which is virtual through technology mediation, and the task space, which is physical [21]. Significant effort has been put into remote collaboration research for replicating co-located settings for remote communication. The challenge is to transmit the various cues that are available in face-to-face, co-located collaboration, between remote and local users.

Additionally, we consider pointing, which is a gesture subset, a non-verbal cue. For audio- and video-mediated collaboration, such as standard video calls, speech cues and cues in the visual channel can be transmitted. However, even with the presence of a video-mediated communication, non-verbal cues can be difficult to transmit due to the viewpoint constraints of cameras in the local scene, as well as the displacement between the physical task space and the video display.

We have identified four interaction primitives needed to support XR Care applications: *pointing*, *gestures*, and *gaze* and *annotations*. These primitives are the building blocks that will be utilized in the construction of all care tasks. While we will draw on previous XR interaction techniques [24,27,31,32,40] in creating these interaction primitives, we will consider specific requirements observed in home care settings, such as: a) the need for the caregiver to have their hands free to execute the task, b) the different bodily orientations that the caregiver and patient need to position themselves in to comfortably guide or perform the tasks, c) the split attention of all the parties due to the tri-partite interaction scenario, d) and positioning of virtual information and control menus in the caregiver space in such a way that the virtual elements do not occlude the caregiver’s vision or impede task execution, nor require the caregiver to gesture or point inappropriately at the patient. These primitives will draw on recent technology advancements in body pose estimation [18,29], object recognition [33,34], gaze tracking [4], and situation awareness modelling [12,36].

Novel Composites. Building on the primitives described, we have identified several novel composite interactions that will enhance stakeholders’ capabilities and perception. We emphasize seemingly “magical interactions” [26] that enhance the capabilities of the local caregiver beyond what is possible in even the fully co-located context. The development of these novel composites is well established for VR (e.g., teleportation and distance grabbing) but has not been explored as much in the context of AR beyond basic annotations [21,32], due to the constraints introduced by the real world [40].

One compelling example for the caregiving context is *replay* and *bookmarking*. *Replay* would give the local caregiver the option to capture and replay segments of procedural instructions. This could

be accomplished by continuously recording all data and positions from all four interaction primitives, and providing the caregiver and clinician a voice or visual interface to bookmark potential segments of interest during their interactive session.

4 NEXT STEPS

To realize this vision for XR as a caregiving tool will require collaboration across many areas of expertise, from XR technologies and systems, to HCI and Interaction design methods. Immediate next steps are to iteratively prototype and study XR Care interaction techniques via both technical evaluation of the interaction components, and user evaluation of exemplar XR Care tasks.

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